Explicit families and limits of degree amplification

2026-11-09. Parallel written derivations and independent review.

Construction or criterion	Established implication
Retain V in a larger field	Same closure at caps at least two
Disjoint satisfiable systems	LFD is the maximum of component LFDs
Slice dual separator of degree c	Slice refutes at cap at most c+2
Polynomial V of degree at most r; n>4r	Exactly two ordered solutions for every r

Explicit family: With z=a6=1, the Semaev equation is $(xy)^2+xy+(x+y+1)^2=0$. No reduction occurs modulo the field polynomial when n>4r. If x and y both have positive polynomial degree, $(xy)^2$ has a unique highest-degree term, a contradiction. Constant substitution leaves precisely (0,1) and (1,0). This removes the existential basis-selection issue for this family.

Degree remains separate: Its r=1 member has exact LFD2 by an explicit affine-ideal derivation. The earlier odd-diagonal lower bound3 does not apply. Solution-count growth or increasing quotient dimension cannot substitute for a full-closure degree obstruction.

Field-specific obligation: For the unique-zero branch H, choose a separator lambda(v) annihilating L_v(W) and pairing to1 with v^2. If its degree is at most c on each coordinate-one slice, then LFD(H)<=c+3. Excluding low-degree separators alone does not exclude arbitrary refutations using u-dependent multipliers. A normalization theorem is missing.

Transport: A degree-s substitution maps source closure at D into target closure at max(sD,c), provided eligible generator images are derived at c. Rank may drop under certificate pullback. Enlarging V additionally requires the named polynomial still to vanish on every new solution. Disjoint products cannot amplify degree.

Evidence: polynomial-subspace-candidate.md and review; field-pencil-dual-obligation.md and review; closure-transport-and-amplification.md and review. No new scientific runs or solver executions. The unbounded-LFD theorem remains open.